

7.PSEUDOCOLOURING

AIM:

The objective of this experiment is to learn how to display grayscale images using pseudocolors in MATLAB.

THEORY:

The purpose of pseudocolor image processing techniques is to enhance a monochrome image for human viewing purposes. Their rationale is that subtle variations of gray levels may very often mask or hide regions of interest within an image. This can be particularly damaging if the masked region is relevant to the application domain (e.g., the presence of a tumor in a medical image). Since the human eye is capable of discerning thousands of color hues and intensities, compared to only less than 100 shades of gray, replacing gray levels with colors leads to better visualization and enhanced capability for detecting relevant details within the image.

INTENSITY SLICING:

The technique of intensity (or density) slicing is the simplest and best-known pseudocoloring technique. If we look at a monochrome image as if it were a 3D plot of gray levels versus spatial coordinates, where the most prominent peaks correspond to the brightest pixels, the technique corresponds to placing several planes parallel to the coordinate plane of the image (also known as the xy plane). Each plane “slices” the 3D function in the area of intersection, resulting in several gray-level intervals. Each side of the plane is then assigned a different color. Figure 1 shows an example of intensity slicing using only one slicing plane at $f(x, y) = I_i$ and Figure 2 shows an alternative representation, where the chosen colors are indicated as c_1, c_2, c_3 , and c_4 . The idea can easily be extended to M planes and $M + 1$ intervals.

CODE:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread(r"C:\Users\Downloads\Fig0107(c)(headCT-Vandy).tif",
cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Could not load image. Please ensure 'Fig0107(c)(headCT-
Vandy).tif' is in the correct directory or uploaded to Colab.")
else:
    m, n = img.shape

    output = np.zeros((m, n, 3), dtype=np.uint8)

    for i in range(m):
        for j in range(n):
            val = img[i, j]
```

```

if 0 <= val < 50:      # Dark Blue
    output[i, j] = [30, 30, 120]

elif 50 <= val < 100:  # Teal
    output[i, j] = [0, 180, 180]

elif 100 <= val < 150: # Violet
    output[i, j] = [150, 50, 200]

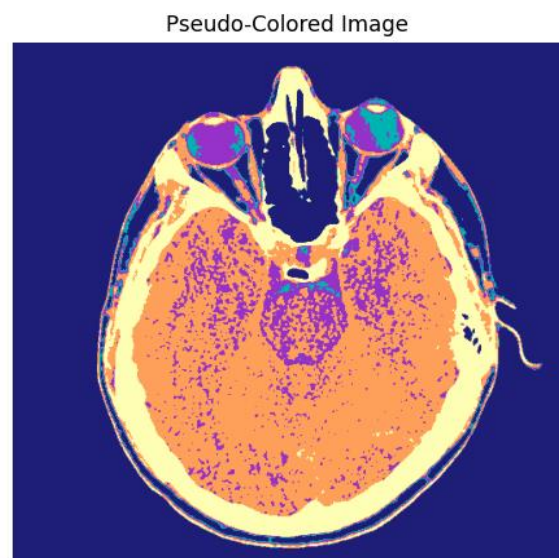
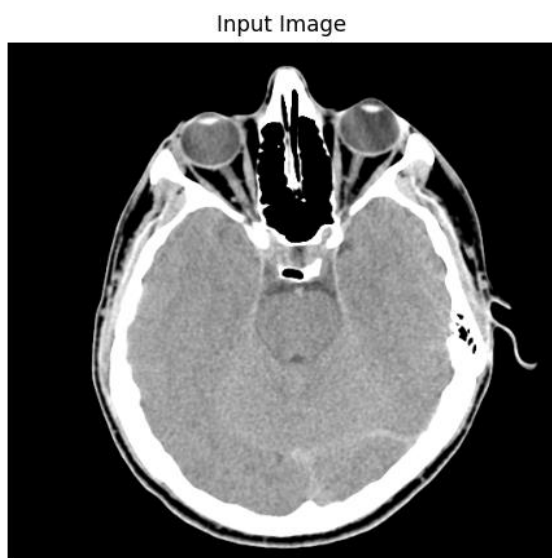
elif 150 <= val < 200: # Peach-Orange
    output[i, j] = [255, 160, 90]

elif 200 <= val < 256: # Pale Yellow-White
    output[i, j] = [255, 255, 180]

plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Input Image")
plt.axis("off")
plt.subplot(1, 2, 2)
plt.imshow(output)
plt.title("Pseudo-Colored Image")
plt.axis("off")
plt.show()

```

OUTPUT:



(Pseudo-coloring enhances the contrast of the CT scan, making anatomical structures and variations more distinguishable than in the grayscale image.)

USING INBUILT FUNCTION

CODE:

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread(r"C:\Users\Downloads\Fig0107(c)(headCT-Vandy).tif",
cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Could not load image.")
else:
    colored = cv2.applyColorMap(img, cv2.COLORMAP_JET)

    plt.figure(figsize=(12, 6))
    plt.subplot(1, 2, 1)
    plt.imshow(img, cmap="gray")
    plt.title("Input Image")
    plt.axis("off")
    plt.subplot(1, 2, 2)
    plt.imshow(cv2.cvtColor(colored, cv2.COLOR_BGR2RGB))
    plt.title("Pseudo-Colored (JET)")
    plt.axis("off")
    plt.show()
```

OUTPUT:



(The JET pseudo-coloring highlights intensity variations in the CT scan, improving visualization of tissue differences for easier interpretation.)

CONCLUSION:

Pseudo-coloring is an effective technique to enhance grayscale medical images for better human interpretation.

It assigns distinct colors to different intensity ranges, making subtle variations more visible.

Intensity slicing provides manual control over mapping gray levels to chosen colors.

Inbuilt functions like `cv2.applyColorMap (JET)` offer fast and standardized pseudo-coloring.

Overall, pseudo-coloring improves visualization of anatomical structures and aids in medical image analysis.